

1

TITLE PLACEHOLDER

2

Paxton C. Fitzpatrick¹, Alishba Tahir^{1,2}, Jennifer Xu¹, and Jeremy R. Manning^{1,*}

¹Department of Psychological and Brain Sciences
Dartmouth College, Hanover, NH 03755, USA

²ALISHBA AFFILIATION PLACEHOLDER
ALISHBA AFFILIATION PLACEHOLDER

*Corresponding author: Jeremy.R.Manning@Dartmouth.edu

3

Abstract

4

ABSTRACT PLACEHOLDER

5

Introduction

6

Suppose that you attend a friend’s wedding. That evening you describe it to your partner, re-
counting how the ceremony ran long, how the best man’s toast was unexpectedly moving, and
how a small child wandered onto the dance floor midway through the first song. A week later
a coworker asks how the wedding was, and you tell the story again. What is that second telling
a telling of? One appealing answer is that both tellings draw on a single underlying memory of
the wedding, and that whatever differences arise between them reflect the ordinary imprecision of
putting an experience into words. A second possibility, which motivates the work we report here,
is that your first telling has itself become part of what you are remembering.

14

What happens to an experience between living it and describing it later? Our brains do not
maintain complete models of everything we encounter, and so the act of perceiving an experience
already entails compressing and distorting it. We attend closely to some aspects of a scene while

16

17 ignoring others, and we tend to interpret ambiguous moments in light of what we expect will
18 happen next. Once the experience has ended, neither the experience itself nor our moment-to-
19 moment perception of it remains available, to us or to anyone else. What remains is the product
20 of a further distillation, in which our perception of the experience is reduced to something our
21 memory systems can store and later reconstruct. Which aspects of an experience survive this second
22 distillation may depend on what we attended to, what we anticipated would matter later, what
23 interested us, and the thoughts, motivations, and biases we brought with us. Because remembering
24 entails reconstructing an experience rather than replaying it, the resulting account often preserves
25 the general shape of what happened while discarding many of its particulars (Bartlett, 1932;
26 Schacter, 2012; Schacter et al., 2011). The units over which these distillations operate appear to be
27 events, in that people segment continuous experience into discrete episodes at moments of change,
28 and those boundaries shape what is remembered afterwards (Baldassano et al., 2017; Radvansky
29 and Zacks, 2017). In prior work we characterized this transformation geometrically, by modeling
30 an experience and a person's recounting of it as trajectories through a semantic space, and asking
31 which properties of the experience's trajectory are preserved in the recounting (Heusser et al.,
32 2021).

33 A memory does not stay fixed once it has been encoded, and this introduces a dimension
34 that a single recounting cannot reveal. Information may be lost over time; experiences we have
35 after an event may color how we later remember it; similar experiences may interfere with one
36 another; and we may simply re-evaluate what happened in light of what we have learned since.
37 Retrieving a memory appears to be one of the forces that changes it, in that a memory can become
38 labile when it is reactivated and must be stabilized again afterwards (Hupbach et al., 2007, 2009;
39 Nader et al., 2000; Sederberg et al., 2011). Retrieval also changes what will be accessible later, often
40 substantially, relative to memories that go unretrieved (Karpicke and Roediger, 2007; Roediger
41 and Karpicke, 2006). The particular way in which we recount an experience matters as well, since
42 people who are asked to describe an event for one purpose later remember it in ways that are biased
43 toward that description (Tversky and Marsh, 2000). Memories also become more semantic and
44 less perceptually detailed, both with the passage of time and with repeated remembering (Lifanov

et al., 2021). Repeated retrieval of naturalistic events over the course of a week has been found to stabilize each person’s recall (for example, which details they include) without causing different people’s recalls to converge (Melega et al., 2025). Taken together, these observations suggest that each time we remember an experience, what we retrieve may differ from what we retrieved before.

One intuition for how this might work comes from thinking about memory storage in terms of attractor dynamics (Hopfield, 1982). Rather than supposing that we hold literal duplicate copies of a memory, we might imagine that laying down a memory carves out a basin in an energy landscape, and that remembering the experience deepens a basin near where we happen to land. Combined with the observation that remembering distorts, this implies that it may not be the original memory that gets re-encoded on each occasion, but rather a distorted version of it. Successive retrievals might then arrive at a basin corresponding to the original experience, or at one corresponding to any of our previous rememberings of it. A related point has been made from the perspective of context reinstatement, namely that each time we remember a prior experience, our memory of that experience becomes colored by the contextual thoughts we were having at the time of each remembering (Manning, 2024). We raise this picture to motivate the questions we ask below rather than as a model we test directly.

How might we compare two rememberings of the same experience, when the experience and both rememberings take the form of unconstrained natural language? Recent work has begun to use language models to quantify the content of what people say when they recount what they remember (Fenerci et al., 2025; Fitzpatrick et al., 2026; Heusser et al., 2021; Manning et al., 2022). Our approach embeds the description of an experience, and each person’s recountings of it, into a common high-dimensional space whose coordinates reflect the concepts under consideration, so that each unfolds over time as a trajectory through that space (Fitzpatrick et al., 2023, 2026; Heusser et al., 2021; Vera et al., 2025). We then segment the experience and each recounting into events, and we identify which event of the experience each recalled event describes, which yields sequences of events that may be compared directly across people and across delays (Baldassano et al., 2017; Chen et al., 2017; Heusser et al., 2021).

Because we collected two recountings of the same experience from each participant, one im-

73 immediately afterwards and one a week later, we are able to ask what the delayed recounting most
74 closely resembles. If retrieval recovered the original experience in an uncorrupted form, then both
75 recountings should be about equally similar to the experience itself. We find instead that a partic-
76 ipant’s delayed recall is more similar to their own immediate recall than it is to the episode. It is
77 also more similar to their own immediate recall than to other participants’ immediate recalls. What
78 governs retrieval after a delay therefore appears to be a person-specific distillation, rather than the
79 episode as it was presented. We report three findings. First, participants recall the episode’s events
80 in a similar order before and after the delay. Second, the semantic structure of a participant’s recall
81 is preserved across the delay, and is specific enough that a delayed recall can be matched to its
82 own immediate recall on the basis of that structure alone. Third, delayed recalls are well described
83 as simplified versions of the same participant’s immediate recalls, in that removing detail from an
84 immediate recall trajectory brings it closer to the delayed recall trajectory that followed it.

85 Results

86 *[OUTLINE — placeholder structure, not final prose. Bracketed italic text marks content still to*
87 *be written. No statistic below has been computed or verified; every “report [STAT]” is a hole to be*
88 *filled from the cited notebook’s output.]*

89 *[FIGURE BUDGET: 8 main-text figures maximum. The plan below uses 6, leaving 2 in re-*
90 *serve (one likely wanted for the RDP analysis if it needs a second panel set, one for contin-*
91 *gency). Figure 1 (**paradigm.pdf**) is the only finished main-text figure; the other PDFs already in*
92 ***paper/figures/source/** are analysis outputs that may feed panels rather than figures in their*
93 *own right. Nothing else is made yet.]*

94 From episode and recalls to comparable event sequences

95 *[Walk the reader through the pipeline, briefly and conceptually (details live in the Methods), up to the point*
96 *where we have three things in hand: (a) the sequence of events comprising the episode, (b) each participant’s*
97 *sequences of immediate and delayed recall events, and (c) the mapping between each recall event and the*

98 episode event it describes. This is the setup that everything below depends on — the point to land is that the
99 episode and both recalls now live in a common space and are segmented into comparable units.]

100 [Figure 1 — **paradigm**. `paradigm.pdf`, already in `paper/figures/source/`. Session-1 row: in-
101 structions, episode (24 min), immediate recall (10–60 min), prediction (5 min); one-week gap; session-2 row:
102 instructions, delayed recall (10–60 min), session-2 episode (20/22 min), recall of that episode (10–60 min).
103 Planned revision: reduce the opacity of the blocks for the phases we don’t analyze — the session-1 prediction
104 block, and the session-2 episode-viewing and subsequent-recall blocks — so the reader’s eye lands on the
105 immediate/delayed recall pair that the paper is actually about.]

106 [Figure 2 — **methods overview**. To be made. Annotations and recall text → trajectories → event-
107 segmentation procedure → event sequences. Loosely modeled on Fig. 2 of Heusser et al. (2021), extended
108 with the (new) event-segmentation step. Layout still open. Candidate source material already generated:
109 `episode-corrmat.pdf` and `k-optimization-episode.pdf` (from `event-segmentation.ipynb`), and
110 the per-recall similarity heatmaps and match matrices `atlep1-and-delayed-recall-similarity-heatmaps.pdf`
111 `/-matchmats.pdf`.]

112 **Participants recall the episode’s events in a similar order before and after the** 113 **delay**

114 [Motivation: we wanted to know whether participants remembered the episode similarly before and after the
115 delay. The simplest way to ask this is whether they recalled the same events in the same order.]

116 [Result: recall-order similarity. Computed in `self-similarity-analyses.ipynb`, section “Is de-
117 layed recall order more similar to own immediate recall order than to others’ immediate recall orders?”. The
118 measure compares pairwise precedence relations over the episode events shared between two recalls, then
119 percentile-ranks each participant’s delayed-vs-own-immediate similarity against their delayed-vs-others’-
120 immediate similarities, tested against chance (0.5) with a Wilcoxon signed-rank test. Report [STAT].]

121 [Figure 3 — **recall-order similarity**. To be made; multi-panel.]

122 [Panel A: illustration of the method. For one example participant, show the matrix of pairwise recall-
123 order-precedence values over pairs of episode events — probably just the lower triangle — for their delayed

124 recall beside their own immediate recall. Grey out (or reduce the opacity of) cells for event pairs not recalled
125 in both sessions, since the comparison is restricted to pairs present in both. Then show the same thing again
126 for that participant's delayed recall against a different participant's immediate recall, which is the contrast
127 the analysis actually tests. This maps directly onto `pairwise_precedence()` and the `shared_events`
128 argument to `order_similarity()` in the notebook, so the panel and the code should be checked against
129 each other.]

130 [Panels B–D: de-slopified versions of panels B–D of the reference mock-up. Include the analogues of that
131 mock-up's B and D so that the same panel types can reappear in the recall-structure figure (Figure 5), letting
132 the reader compare order and structure on identical terms — this parallelism is the reason for the pairing
133 and should be preserved in the final layout.]

134 [Note on ordering: this is the least novel analysis of the three and the one least dependent on our
135 framework — order could be measured without any embedding model. It goes first precisely because it
136 establishes the basic effect on the simplest possible terms, setting up the contrast with what follows.]

137 **The semantic structure of participants' recalls is preserved across the delay**

138 [Motivation: order is only the coarsest description of a recall. Because our approach embeds recalls in a
139 continuous semantic space, we can ask a stronger question — whether the structure of how a participant
140 moved through the episode's content is preserved, not merely the sequence in which they hit the events.]

141 [Figure 4 (optional) — 2D trajectories. Decision deferred, and it depends on how this section's
142 narrative settles. Argument for including it: it introduces the idea of geometric comparison and establishes
143 the 2D paths before they get used as examples in Figure 5, so the reader isn't meeting the representation
144 and the result at the same time. Argument against: it spends one of 8 figure slots on setup. Middle
145 option — a compact version showing only the episode path plus immediate and delayed paths for a handful
146 of participants, with the full version relegated to the supplement. Source: `2d-trajectories.ipynb`
147 (`atlep1-paths2d.pdf` and `delayed-paths2d.pdf`, `imm-del-2d.pdf`); note those `savefig` calls are
148 currently commented out in the notebook.]

149 [Result: trajectory/event-structure similarity. Computed in `self-similarity-analyses.ipynb`,
150 sections "Is delayed recall more similar to own immediate recall than to episode?" and "Is delayed recall

151 *more similar to own immediate recall than to others' immediate recalls?''*. Both use DTW warp cost (average
152 *cosine distance between matched events*) over mean-centered event sequences; the second percentile-ranks
153 *self against others and tests against chance. Report [STAT] for each.*]

154 *[Framing: this result appears strong enough to present as a "fingerprint" — i.e., that one can identify*
155 *which immediate recall belongs to a given participant from the structure of their delayed recall alone. Frame*
156 *it that way in both the text and the figure, but confirm the effect size supports the claim before committing*
157 *to that language.]*

158 **[Figure 5 — recall-structure similarity.** *To be made; the largest figure in the paper, covering both*
159 *analysis (A) "delayed is more similar to immediate than to the episode" and analysis (B) "delayed is more*
160 *similar to own immediate than to others' immediates". Layout still open.]*

161 *[Example trajectory panels (3, from the current draft): the left panel is common to both analyses, so*
162 *it should sit centrally in the final arrangement; the right panel is the contrast for analysis (A) and the*
163 *middle panel is the contrast for analysis (B). Planned refinements: thicker lines; faint connectors between*
164 *DTW-matched events within each panel (already present in the draft); possibly PCHIP splines rather than*
165 *straight segments between points, to make the paths read as curves. Under consideration — a fourth panel*
166 *showing pairwise mean point-by-point distances between trajectories, which would require upsampling each*
167 *trajectory to a common number of points and comparing them in order (or substituting DTW cost, if that*
168 *turns out to read better).]*

169 *[Analysis (A) panel: per-participant connected dots showing each participant's delayed-to-immediate*
170 *versus delayed-to-episode alignment cost. Probably without the KDEs from the draft version.]*

171 *[Analysis (B) panels (3): de-slopified versions of the current draft. Two of them are the direct analogues*
172 *of Figure 3's panels B and D and show stronger effects — keeping the panel types visually identical across*
173 *the two figures is what makes "structure beats order" legible without the reader doing arithmetic. The third*
174 *gets at how person-specific the similarity is, and is the panel that carries the "fingerprint" claim.]*

175 *[Candidate supporting material: `atlep1-recall-corrstats.pdf` and `delayed-recall-corrstats.pdf`*
176 *(from `event-segmentation.ipynb`).]*

177 **Delayed recalls are simplified versions of participants' immediate recalls**

178 *[Motivation: having established that a participant's delayed recall structure resembles their own immediate*
179 *recall structure, the natural next question is how it changes over the delay.]*

180 *[Result: trajectory simplification. Computed in `rdp-analyses.ipynb`. Each participant's immediate*
181 *recall trajectory is simplified with the Ramer-Douglas-Peucker algorithm over a range of tolerances (ϵ), and*
182 *the ϵ minimizing DTW warp cost to their delayed recall trajectory is selected. Sub-results to report: (i) the*
183 *relationship between how many events a participant dropped across the delay and how many RDP removes*
184 *at the best-fitting ϵ (Kendall's τ); (ii) the same against ϵ itself; (iii) whether RDP predicts which events*
185 *were forgotten, benchmarked against a hypergeometric null and compared to predictions derived from other*
186 *participants' trajectories. Report [STAT] for each.]*

187 *[Figure 6 — RDP simplification. To be made, and how best to illustrate the simplification is genuinely*
188 *unresolved — flagged as an open design question rather than a layout still to be drawn. The regression plots*
189 *for sub-results (i) and (ii) exist in `rdp-analyses.ipynb` but have no `savefig` calls, so nothing is exported*
190 *to `paper/figures/source` yet. One thing worth trying: since Figure 5 will already have taught the reader*
191 *to read 2D trajectories, showing one participant's immediate path beside its RDP-simplified version and*
192 *their actual delayed path would make “the delayed recall looks like a simplified version of the immediate one”*
193 *visible directly, with the quantitative panels as support. If the analysis needs a second panel set, one of the 2*
194 *reserve figure slots can absorb it.]*

195 *[@paxton: does the simplification story want a fourth subsection on what the dropped events have in*
196 *common (e.g. are they lower-salience, more peripheral to the narrative), or is that a Discussion point?]*

197 **Discussion**

198 **Methods**

199 Our experimental protocol was approved by the Committee for the Protection of Human Subjects
200 at Dartmouth College. We obtained written informed consent from all participants prior to the
201 start of both experiment sessions.

202 **Participants**

203 We enrolled a total of 70 Dartmouth undergraduate students in our study. Participants received
204 optional course credit for enrolling and were pre-screened to ensure they had not previously seen
205 any episode of either TV show viewed during the experiment. No statistical method was used
206 to predetermine sample size. Prior to data processing and analysis, we excluded a total of 17
207 participants from our dataset: 4 encountered technical issues with the experiment, 2 self-reported
208 difficulty with the task due to illness, 4 spoke for less than 10 minutes during the immediate recall
209 phase, and 7 did not return for the second experiment session. The data we analyzed comprised
210 the remaining 53 participants.

211 At the start of the experiment, we asked each participant to complete a demographic survey
212 that included questions about their age, sex, race, and ethnicity. Participants' ages ranged from 18
213 to 22 years (mean: 18.91 years; standard deviation: 1.01 years). A total of 41 participants reported
214 their sex as female and 12 participants reported their sex as male. Participants reported their races
215 as White (34 participants), Asian (16 participants), Black or African American (3 participants),
216 American Indian or Alaska Native (1 participant), and other (1 participant). (Note that some
217 participants selected multiple racial categories.) A total of 52 participants reported their ethnicity
218 as "Not Hispanic or Latino" and 1 participant reported their ethnicity as "Hispanic or Latino."

219 We also asked participants their native spoken language; whether they had any hearing, speech,
220 or color vision impairments; and about any current medications or recent injuries that could impact
221 attention or memory. All participants reported that their native spoken language was English, that
222 they had no hearing or speech impairments, and that they had normal color vision. A total of 52
223 participants reported no attention- or memory-relevant medications or injuries. One participant
224 reported sustaining a concussion more than 3 years prior to the experiment and was not excluded
225 from our dataset.

226 Additionally, before each of the two experiment sessions, we asked participants about their
227 sleep, coffee consumption, and level of alertness. Participants reported having slept between 5
228 and 11 hours the night prior to session 1 (mean: 7.37 hours; standard deviation: 1.45 hours) and

between 5 and 11 hours the night prior to session 2 (mean: 7.39 hours; standard deviation: 1.23 hours). They reported having consumed between 0 and 3 cups of coffee on the day of session 1 (mean: 0.43 cups; standard deviation: 0.72 cups) and between 0 and 3 cups of coffee on the day of session 2 (mean: 0.48 cups; standard deviation: 0.68 cups). Participants reported their current levels of alertness on a 5-point Likert scale. At the start of session 1, 1 participant reported feeling "Very sluggish," 14 reported feeling "A little sluggish," 18 reported feeling "Neutral," 15 reported feeling "A little alert," and 5 reported feeling "Very alert." At the start of session 2, 3 participants reported feeling "Very sluggish," 9 reported feeling "A little sluggish," 23 reported feeling "Neutral," 17 reported feeling "A little alert," and 3 reported feeling "Very alert."

@jeremy: do we want to report class years and majors? And/or do we want to report post-experiment survey responses (how engaging did you find the episode, how easy/difficult was it to follow, how well do you think you recalled it, how well do you think you learned characters' names)?

Experiment

The experiment took place over two sessions lasting approximately one hour each, separated by 6–8 days (mean: 6.76 days; standard deviation: 0.73 days). Prior to the start of the experiment, participants were assigned to one of two groups (*A* or *B*), alternating based on enrollment order. During session 1, all participants first viewed the pilot episode of the FX series *Atlanta* ("The Big Bang"; duration: 24 min 15 s). Immediately after viewing, participants were given up to one hour to recount what happened in the episode to the best of their abilities. Following Chen et al. (2017), participants were instructed to describe the episode in as much detail as possible and to try to speak for a minimum of 10 minutes. They were told to try to recount the episode's events in their original order, but that they were free to go back and describe earlier events should they realize they had missed something. After the immediate recall phase, participants were then given up to 5 minutes to predict what they thought would happen in the show's next episode.

Upon returning for session 2, all participants were again given up to one hour to recount the *Atlanta* episode they had viewed the week prior. Following the delayed recall phase, participants in Group *A* then viewed the second episode of *Atlanta* ("Streets on Lock"; duration: 21 min

57 s) while participants in Group B instead viewed the pilot episode of the FOX series *Arrested Development* (“Pilot”; duration: 20 min 37 s). Both groups were then given up to one hour to recount the just-watched episode in the same manner and given the same instructions as in session 1. We implemented the experiment using the psiTurk platform (Gureckis et al., 2016) and jsPsych library (de Leeuw, 2015), with a custom jsPsych plugin we developed in prior work for recording verbal recall audio (Ziman et al., 2018). Here we analyze participants’ immediate and delayed recalls of the *Atlanta* episode viewed in session 1.

Stimulus annotation and recall transcription

To capture the evolving semantic content of the *Atlanta* episode, we created detailed text annotations of the on-screen content during each individual camera shot throughout the episode (mean annotation duration: 3.05 s; standard deviation: 1.80 s). Specifically, we used ANVIL (Kipp, 2001) to record, for each camera shot, a set of semantically relevant features. These included a brief description of physical occurrences and characters’ explicit actions, a brief description of characters’ internal states (e.g., emotions, motivations) and other implicit content, the names of all characters appearing on screen, a transcription of any dialogue, the name(s) of the character(s) speaking, the shot’s setting, whether the shot took place indoors or outdoors, and whether or not there was music playing during the shot. This feature set was adapted from a similar annotation corpus created by Chen et al. (2017), which we used in developing our trajectory-analysis framework in prior work (Heusser et al., 2021).

We then transcribed each participant’s immediate and delayed verbal recall of the *Atlanta* episode. Our transcription process consisted of three steps: an initial manual transcription, a two-pass prose-formatting conversion with a large language model, and a final manual cleanup and correction step. First, we manually transcribed each recall audio file collected during the experiment. These initial transcripts reflected participants’ spoken recalls verbatim, without any punctuation or additional formatting, and including disfluencies, stuttered repetitions, and other artifacts common to unrehearsed speech. However, since transformer models (such as we used to obtain text embeddings of participants’ recalls) are trained almost exclusively on written text,

structural conventions like sentence breaks, punctuation, and capitalization provide an important signal these models leverage to contextual other tokens (e.g., words) and construct a representation of the text’s overall meaning. Meanwhile, speech-native phenomena such as stuttered repetitions and filler words (“like”, “um”, etc.) that are comparatively rare in written text contribute out-of-distribution noise that can be expected to degrade embedding fidelity, and prior work has shown that automated removal of these phenomena improves performance on downstream tasks that assume clean, written-style input (Chen et al., 2022; Wang et al., 2010). We therefore used a large language model (Qwen3.5 122b; Qwen Team, 2026) to convert participants’ recall transcripts to a structure and style more typical of written text. We used the `langchain_dartmouth` Python package (Stone et al., 2025) to implement this conversion as two sequential passes over each transcript. In the first pass, we prompted the model to split the transcript into sentences at natural breaking points, add punctuation including quotes around reported speech, remove clear non-word disfluencies (“um,” “uh,” “hmm,” etc.), collapse stuttered verbatim repetitions, and mark self-corrections and restarts with em-dashes. In the second pass, we prompted the model to remove all instances of “like” used as a filler word. We removed filler “like” instances in a separate pass because of their high frequency of occurrence in the transcripts and the additional effort to required to distinguish them from meaningful uses of “like.” The full texts of our prompts for both passes can be found in *Supplementary methods*. Finally, we manually reviewed all LLM-formatted transcripts to identify and correct any errors, ensure sentence breaks were appropriate and consistent across participants, confirm that no content was improperly added to or removed from the transcripts, and resolve any ambiguities based on the prosody of participants’ original recall audio.

Analysis

Statistics

Wilcoxon tests were one-sided because our alternative hypothesis was directional (delayed matches own immediate *better than* the episode, others’ immediates, etc.).

309 Constructing episode and recall trajectories

310 We adapted a method we developed in prior work (Heusser et al., 2021) to transform the *Atlanta*
311 episode and participant’s recalls of it into trajectories through a common representation space
312 using a text embedding model. Briefly, text embedding models take in natural language text
313 of an arbitrary length and output a fixed-length vector representation that captures the text’s
314 semantic meaning. Intuitively, an output vector of length n can be viewed as a coordinate in an
315 n -dimensional text embedding space. While specific model architectures and training paradigms
316 vary, modern text embedding models are broadly trained to output similar coordinate vectors (in
317 terms of cosine similarity, dot product, inverse Euclidean distance, or other geometric measures) for
318 semantically similar input texts, even when those texts share few or no specific words. We obtained
319 text embeddings for the episode and recalls using EmbeddingGemma (Vera et al., 2025), a pre-
320 trained 308-million-parameter encoder-only transformer model that outputs coordinate vectors in
321 a 768-dimensional embedding space.

322 To construct the trajectory for the *Atlanta* episode, we first converted our tabular annotations
323 of the content present during each camera shot to a plain-text format. Separately for each shot,
324 we concatenated the annotation text describing explicit on-screen occurrences and characters’
325 implicit internal states. We then appended “Speech: ” followed by the shot’s dialogue, “Character
326 speaking: ” followed by the name of the speaker, and “Setting: ” followed by the shot’s setting. If
327 any annotated fields were empty for a given shot (e.g., if no character was speaking at that time), we
328 omitted them and their prefixes from the concatenated text. Next, we used EmbeddingGemma to
329 transform the text snippet for each camera shot into a text embedding vector, resulting in a number-
330 of-shots (478) by number-of-features (768) matrix. Since the camera shots varied in duration, we
331 then resampled this series of embedding vectors to a consistent temporal resolution. We assigned
332 each vector to the timepoint midway between the start and end of its corresponding camera shot
333 (in seconds), and used linear interpolation to estimate a vector for each second of the episode. This
334 yielded a number-of-seconds (1,455) by number-of-features (768) matrix—i.e., a 768-dimensional
335 *trajectory*.

336 We then created similar trajectories for each participant’s immediate and delayed recall of the
337 episode. We began by splitting each recall transcript into a list of sentences, with the constraint
338 that splits could not occur within reported speech (i.e., if a participant recounted a multi-sentence
339 quote by a character in the episode, we treated the full quote as a single sentence-unit). We
340 then parsed these sentences into overlapping sliding windows that spanned up to 5 consecutive
341 sentences each. These windows grew to and shrank from their full size at the beginning and end of
342 a participant’s recall sequence—in other words, the first window contained only the first sentence
343 of the participant’s recall, the second contained the first two sentences, and so on. This ensured
344 that all sentences from a participant’s recall transcript appeared an equal number of times (5) in
345 their sequence of sliding windows. Finally, we used EmbeddingGemma to transform each sliding
346 window into a text embedding vector. Repeated for all recall transcripts, this yielded a separate
347 trajectory characterizing each participant’s recall from each of the two experiment sessions. The
348 number of samples in participants’ recall trajectories (i.e., the number of sliding windows parsed
349 from their recall transcripts) ranged from 61 to 308 (mean: 136.68; standard deviation: 49.70).

350 **Relating episode and recall embedding vectors**

351 In prior work using our trajectory-analysis framework (e.g., Fitzpatrick et al., 2026; Heusser et al.,
352 2021; Manning, 2021; Manning et al., 2022; Racciah et al., 2024), embeddings of the stimulus and
353 participants’ responses were constructed using a topic model (Latent Dirichlet Allocation; Blei
354 et al., 2003) trained solely on the stimulus content. Here we elected to instead use a pre-trained
355 transformer-based text embedding model due to the richer and more nuanced textual information
356 these models are able to capture. (For example, these context-sensitive models can disambiguate
357 the texts “Earn looked at Alfred” and “Alfred looked at Earn” in their representations, whereas
358 bag-of-words models like LDA would represent the two identically.) However, these richer rep-
359 resentations also introduce certain geometric nuisances we needed to account for in order to
360 effectively relate the contents of participants’ recalls to that of the episode they viewed. First,
361 because modern transformer models are pre-trained on large and diverse text corpora, the repre-
362 sentational capacities of their embedding spaces span an enormous breadth of semantic content.

363 Consequently, in preliminary analyses we found that the embeddings of our annotations and par-
364 ticipants’ recalls (which were all “about” the events, themes, and characters of a single *Atlanta*
365 episode) were strongly anisotropic—i.e., they all occupied the same relatively narrow angular
366 region of the model’s overall embedding space. Additionally, because transformer models are sen-
367 sitive to the syntactic structure of text, their embedding representations carry latent information
368 about the text’s linguistic register in addition to its semantic content. Accordingly, we found there
369 was a consistent angular offset between embeddings of the episode annotations and those of par-
370 ticipants’ recalls, reflecting the fact that the annotations comprise grammatically well-structured
371 written text while the recalls comprise transcripts of casual unrehearsed speech.

372 We corrected for these representational artifacts by mean-centering the episode and recall trajec-
373 tories before performing analyses that involved measuring similarities between embedding vectors
374 within or across trajectories. Conceptually, mean-centering a given set of embedding vectors (i.e.,
375 subtracting the across-vector average from each individual vector) serves to remove representa-
376 tions that are common across all vectors under consideration, and instead highlight representations
377 that *vary* between them. For example, consider the sequence of embedding vectors comprising
378 the trajectory of the *Atlanta* episode. Common across vectors within this sequence will be latent
379 signals driven by how we chose to format the annotation text passed to the embedding model, the
380 annotator’s particular idiolect (i.e., idiosyncratic vocabulary and writing style), and any high-level
381 overarching themes of the episode. *Not* common across these vectors are the finer-grained semantic
382 representations that distinguish different events, scenes, and actions throughout the episode from
383 each other. We found that removing these non-discriminative signals (to the extent they behave as
384 additive offsets) via mean-centering helped reduce the embeddings’ anisotropy (i.e., increased the
385 range of measured between-vector similarities) and improved our ability to segment the episode
386 trajectory into meaningful events. Similarly, mean-centering both the episode and recalls before
387 relating their embedding representations helped mitigate broad-scale representational differences
388 driven by their diverging registers, and enabled us to better relate them based on event-specific,
389 content-driven patterns.

390 Our approach to mean-centering the recalls was guided by the particular latent representation

we wanted to remove versus retain. In segmenting each participant’s recall trajectory from each session into events (see *Event-segmentation*), our goal was to discriminate between embedding vectors strictly within a single recall sequence in order to identify temporally transient content-related patterns. We therefore centered each recall trajectory (in addition to the episode) on its own centroid to remove non-discriminative representations (e.g., register, broad-scale thematic content) and maximize the contrast between vectors within that single sequence. However, the analyses shown in Figures **REF TO DTW FIG** and **REF TO RDP FIG** entailed comparing sequences of recall events not only to the episode, but also to each other, both within- and across-participants, to examine whether within-participant similarities tended to exceed across-participant similarities. Here our goal was to remove representational artifacts that could work to bias our results towards significance. Specifically, participants’ idiolects (i.e., particular speaking styles and sets of words with which they describe the episode) will influence the embedding representations of their recalls in ways that tend to be similar between their immediate and delayed recalls, but different across participants. To correct for this confound, we jointly centered each participant’s set of immediate and delayed recall events on the two sequences’ shared centroid (i.e., the average vector across all events from both sessions). This served to reduce the recall embeddings’ anisotropy (by removing signals driven by high-level themes of the episode they recalled) and mitigate the effects of the spoken-recall register and participant-specific idiolect (both of which are shared between a participant’s recalls from the two sessions) while preserving any genuine drift in the high-level content of each participant’s recall over the delay (which per-session mean-centering would instead discard).

Event-segmentation

After constructing trajectories for the episode and each participant’s immediate and delayed recall, we used hidden Markov models (HMMs; Baum and Petrie, 1966; Rabiner, 1989) to segment each trajectory into a sequence of discrete events. Given a number of states K , an HMM models an observed time series as a progression through K hidden states, each of which has a characteristic signature in the data (i.e., a pattern that timepoints belonging to that state tend to exhibit) and

tends to persist across timepoints with occasional transitions rather than fluctuating freely. Because states persist, inferring the most probable state at each timepoint segments the observed time series into a sequence of events—i.e., spans of contiguous timepoints with the same most-probable state.

We segmented the episode trajectory into events using the HMM implemented in the BrainIAK toolbox (Kumar et al., 2021), which jointly learns a mean pattern for each of K states and the set of transitions between them via the Baum-Welch algorithm (Baum et al., 1970), with the added constraint that each state is visited exactly once in a strict left-to-right order (Baldassano et al., 2017). To select an appropriate number of states (i.e., events) for the episode, we used an optimization procedure we developed in prior work (Heusser et al., 2021) that aims to identify the single timescale of event representations emphasized most heavily in the episode’s narrative structure. We began by segmenting the (mean-centered) episode trajectory into K events for each K between 2 and 50, inclusive. For each candidate segmentation, we then computed the cosine similarity between each pair of timepoints assigned to the same event, and between each pair of timepoints assigned to different events. When constructing these distributions of within- and across-event similarities, we restricted comparisons to pairs of timepoints separated by no more than the duration of the longest event in the candidate segmentation, so that the within- and across-event distributions were computed over comparable temporal distances and the across-event distribution was not dominated by distant timepoint pairs that we would expect to be dissimilar regardless of event structure. We then selected the candidate segmentation that maximized the Wasserstein distance between these within- and across-event similarity distributions. We found that the maximally distinctive and internally self-similar segmentation was yielded by $K = 20$ events. **(TODO: ADD SUPP FIG SHOWING WASSERSTEIN DIST AS FUNCTION OF K)** Finally, we averaged the embedding vectors over timepoints between each pair of adjacent event boundaries to obtain a sequence of stable event-average representations for the episode.

To segment participants’ recall trajectories into sequences of events, we developed a novel HMM-driven procedure that grounded the recalls directly against the content of the episode participants viewed and recounted. Essentially, rather than re-learning separate sets of state representations from each participant’s recalls (which tend to be shorter, noisier approximations

446 of the episode content), we leveraged the state representations already learned from the episode to
 447 identify transitions between the episode events a given participant was (most probably) describing
 448 at a given point in time. First, we mean-centered the sequence of event-average embedding
 449 vectors for the episode, and each participant’s sequence of per-window embedding vectors from
 450 each session, separately. For each given recall trajectory, we then computed the cosine similarity
 451 between the embedding vector for each recall window and the event-average embedding vector
 452 for each of the 20 episode events. This yielded a number-of-windows by number-of-episode-
 453 events (20) similarity matrix. We then treated the episode events as the hidden states of a hidden
 454 Markov model with fixed parameters, and treated each recall window’s profile of similarities
 455 to the episode events, z-scored across all window-event pairs, as its distribution of emission
 456 scores over states. We imposed an ergodic transition structure (i.e., we allowed transitions from
 457 any state to any other state, since participants may recall the episode events out of order, and
 458 allowed states to be re-entered multiple times, since participants may revisit earlier events in
 459 their recalls) governed by separate additive transition penalties for forward-one transitions and
 460 all other transitions. Specifically, we imposed no penalty for remaining in the current state, a
 461 small penalty (1.5 normalized emissions units) for transitioning from a given state to the next
 462 (i.e., transitioning from recalling episode event n to recalling episode event $n + 1$), and a larger
 463 penalty (3.0 normalized emissions units) for all other transitions. Finally, we decoded the most
 464 probable sequence of episode-event labels using the Viterbi algorithm (Viterbi, 1967), collapsed
 465 runs of consecutive identically labeled recall windows into events, and averaged the embedding
 466 vectors over windows within each event. This procedure yielded, for each participant’s recall from
 467 each session, a sequence of event-average embedding vectors and a set of labels denoting which
 468 episode event each recall event corresponded to.

469 **Comparing trajectory structures**

470 We compared the geometric structures of the episode and recall event-sequences using dynamic
 471 time warping (DTW; Sakoe and Chiba, 1978; Vintsyuk, 1968). The DTW algorithm finds the optimal
 472 order-preserving alignment between two sequences of vectors with potentially different lengths or

sampling rates by locally “stretching” and “compressing” them in time to maximize their overall similarity. This alignment yields a “warp path” that maps each vector in each sequence onto one or more consecutive vectors in the other sequence with the minimum total distance between matched vectors. The summed or average distance between matched vectors along the warp path (the “warp cost”) is commonly taken as a measure of the two sequences’ structural dissimilarity. We computed the warp cost as the average cosine distance between DTW-matched events so that the cost values would be comparable across comparisons involving event-sequences of different lengths (whereas summed distance necessarily grows with sequence length).

Data availability

All of the data analyzed in this manuscript may be found at <https://github.com/ContextLab/memory-dynamics>.

Code availability

All of the code for running our experiment and carrying out the analyses may be found at <https://github.com/ContextLab/memory-dynamics>.

References

- Baldassano, C., Chen, J., Zadbood, A., Pillow, J. W., Hasson, U., and Norman, K. A. (2017). Discovering event structure in continuous narrative perception and memory. *Neuron*, 95(3):709–721.
- Bartlett, F. C. (1932). *Remembering: A study in experimental and social psychology*. Oxford University Press.
- Baum, L. E. and Petrie, T. (1966). Statistical inference for probabilistic functions of finite state Markov chains. *Annals of Mathematical Statistics*, 37(6):1554–1563.

- 495 Baum, L. E., Petrie, T., Soules, G., and Weiss, N. (1970). A maximization technique occurring in the
 496 statistical analysis of probabilistic functions of Markov chains. *Annals of Mathematical Statistics*,
 497 41(1):164–171.
- 498 Blei, D. M., Ng, A. Y., and Jordan, M. I. (2003). Latent dirichlet allocation. *Journal of Machine*
 499 *Learning Research*, 3:993–1022.
- 500 Chen, A., Zayats, V., Walker, D., and Padfield, D. (2022). Teaching BERT to wait: balancing accuracy
 501 and latency for streaming disfluency detection. In Carpuat, M., de Marneffe, M. C., and Ruiz,
 502 I. V. M., editors, *Proceedings of the 2022 Conference of the North American Chapter of the Association*
 503 *for Computational Linguistics: Human Language Technologies*, pages 827–838, Seattle, United States.
 504 Association for Computational Linguistics.
- 505 Chen, J., Leong, Y. C., Honey, C. J., Yong, C. H., Norman, K. A., and Hasson, U. (2017). Shared
 506 memories reveal shared structure in neural activity across individuals. *Nature Neuroscience*,
 507 20(1):115.
- 508 de Leeuw, J. R. (2015). jsPsych: A JavaScript library for creating behavioral experiments in a web
 509 browser. *Behavior Research Methods*, 47(1):1–12.
- 510 Fenerci, C., Cheng, Z., Addis, D. R., Bellana, B., and Sheldon, S. (2025). Studying memory narratives
 511 with natural language processing. *Trends in Cognitive Sciences*, 29(6):516–525.
- 512 Fitzpatrick, P. C., Heusser, A. C., and Manning, J. R. (2023). Text embedding models yield high-
 513 resolution insights into conceptual knowledge from short multiple-choice quizzes. *PsyArXiv*.
- 514 Fitzpatrick, P. C., Heusser, A. C., and Manning, J. R. (2026). Text embedding models yield detailed
 515 conceptual knowledge maps derived from short multiple-choice quizzes. *Nature Communications*,
 516 17(1):2955.
- 517 Gureckis, T. M., Martin, J., McDonnell, J., Rich, A. S., Markant, D., Coenen, A., Halpern, D.,
 518 Hamrick, J. B., and Chan, P. (2016). psiTurk: an open-source framework for conducting replicable
 519 behavioral experiments online. *Behavior Research Methods*, 48(3):829–842.

- 520 Heusser, A. C., Fitzpatrick, P. C., and Manning, J. R. (2021). Geometric models reveal behavioural
521 and neural signatures of transforming experiences into memories. *Nature Human Behaviour*,
522 5(7):905–919.
- 523 Hopfield, J. J. (1982). Neural networks and physical systems with emergent collective computa-
524 tional abilities. *Proceedings of the National Academy of Sciences, USA*, 79:2554–2558.
- 525 Hubbach, A., Gomez, R., Hardt, O., and Nadel, L. (2007). Reconsolidation of episodic memories:
526 a subtle reminder triggers integration of new information. *Learning and Memory*, 14:47–53.
- 527 Hubbach, A., Gomez, R., and Nadel, L. (2009). Episodic memory reconsolidation: updating or
528 source confusion? *Memory*, 17(5):502–510.
- 529 Karpicke, J. D. and Roediger, H. L. (2007). Repeated retrieval during learning is the key to long-term
530 retention. *Journal of Memory and Language*, 57:151–162.
- 531 Kipp, M. (2001). Anvil-a generic annotation tool for multimodal dialogue. In *Proceedings of the*
532 *Conference on Speech Communication and Technology*.
- 533 Kumar, M., Anderson, M. J., Antony, J. W., Baldassano, C., Brooks, P. P., Cai, M. B., Chen, P.-H.,
534 Ellis, C. T., Henselman-Petrusek, G., Huberdeau, D., Hutchinson, J. B., Li, Y. P., Lu, Q., Manning,
535 J. R., Mennen, A. C., Nastase, S. A., Richard, H., Schapiro, A. C., Schuck, N. W., Shvartsman, M.,
536 Sundaram, N., Suo, D., Turek, J. S., Turner, D., Vo, V. A., Wallace, G., Wang, Y., Williams, J. A.,
537 Zhang, H., Zhu, X., Capota, M., Cohen, J. D., Hasson, U., Li, K., Ramadge, P. J., Turk-Browne,
538 N. B., Willke, T. L., and Norman, K. A. (2021). BrainIAK: the brain image analysis kit. *Aperture*,
539 pages doi.org/10.52294/31bb5b68–2184–411b–8c00–a1dadb61e1da.
- 540 Lifanov, J., Linde-Domingo, J., and Wimber, M. (2021). Feature-specific reaction times reveal a
541 semanticisation of memories over time and with repeated remembering. *Nature Communications*,
542 12(1):3177.
- 543 Manning, J. R. (2021). Episodic memory: mental time travel or a quantum “memory wave”
544 function? *Psychological Review*, 128(4):711–725.

545 Manning, J. R. (2024). Context reinstatement. In Kahana, M. J. and Wagner, A. D., editors, *The*
546 *Oxford Handbook of Human Memory*. Oxford University Press.

547 Manning, J. R., Notaro, G. M., Chen, E., and Fitzpatrick, P. C. (2022). Fitness tracking reveals task-
548 specific associations between memory, mental health, and physical activity. *Scientific Reports*,
549 12(1):13822.

550 Melega, G., Samson, K., Lee, H., and Renoult, L. (2025). The semantic structure of events consis-
551 tently influences episodic memory recall over time in young and older adults. *Scientific Reports*,
552 15(1):41987.

553 Nader, K., Schafe, G. E., and LeDoux, J. E. (2000). Fear memories require protein synthesis in the
554 amygdala for reconsolidation after retrieval. *Nature*, 406(6797):722–726.

555 Qwen Team (2026). Qwen3.5: towards native multimodal agents. [https://qwen.ai/blog?id=](https://qwen.ai/blog?id=qwen3.5)
556 [qwen3.5](https://qwen.ai/blog?id=qwen3.5).

557 Rabiner, L. R. (1989). A tutorial on Hidden Markov Models and selected applications in speech
558 recognition. *Proceedings of the IEEE*, 77(2):257–286.

559 Raccach, O., Chen, P., Gureckis, T. M., Poeppel, D., and Vo, V. A. (2024). The “Naturalistic Free
560 Recall” dataset: four stories, hundreds of participants, and high-fidelity transcriptions. *Scientific*
561 *Data*, 11(1):1317.

562 Radvansky, G. A. and Zacks, J. M. (2017). Event boundaries in memory and cognition. *Current*
563 *Opinion in Behavioral Sciences*, 17:133–140.

564 Roediger, H. L. and Karpicke, J. D. (2006). Test-enhanced learning. *Psychological Science*, 17(3):249–
565 255.

566 Sakoe, H. and Chiba, S. (1978). Dynamic programming algorithm optimization for spoken word
567 recognition. *IEEE Transactions on Acoustics, Speech, and Signal Processing*, 26(1):43–49.

568 Schacter, D. L. (2012). Constructive memory: past and future. *Dialogues in Clinical Neurosciences*,
569 1:7–18.

570 Schacter, D. L., Guerin, S. A., and St Jacques, P. L. (2011). Memory distortion: an adaptive
571 perspective. *Trends in Cognitive Sciences*, 15(10):467–474.

572 Sederberg, P. B., Gershman, S. J., Polyn, S. M., and Norman, K. A. (2011). Human memory
573 reconsolidation can be explained using the temporal context model. *Psychonomic Bulletin and*
574 *Review*, 18(3):455–468.

575 Stone, S., Crossett, J., Luker, T., Leligdon, L., Cowen, W., and Darabos, C. (2025). Dartmouth Chat
576 - deploying an open-source LLM stack at scale. In *Practice and Experience in Advanced Research*
577 *Computing 2025: the Power of Collaboration*, pages 1–5, New York, NY, USA. Association for
578 Computing Machinery.

579 Tversky, B. and Marsh, E. J. (2000). Biased retelling of events yields biased memories. *Cognitive*
580 *Psychology*, 40:1–38.

581 Vera, H. S., Dua, S., Zhang, B., Salz, D., Mullins, R., Panyam, S. R., Smoot, S., Naim, I., Zou, J.,
582 Chen, F., Cer, D., Lisak, A., Choi, M., Gonzalez, L., Sanseviero, O., Cameron, G., Ballantyne, I.,
583 Black, K., Chen, K., Wang, W., Li, Z., Martins, G., Lee, J., Sherwood, M., Ji, J., Wu, R., Zheng,
584 J., Singh, J., Sharma, A., Sreepathihalli, D., Jain, A., Elarabawy, A., Co, A., Doumanoglou, A.,
585 Samari, B., Hora, B., Potetz, B., Kim, D., Alfonseca, E., Moiseev, F., Han, F., Gomez, F. P., Ábrego,
586 G. H., Zhang, H., Hui, H., Han, J., Gill, K., Chen, K., Chen, K., Shanbhogue, M., Boratko, M.,
587 Suganthan, P., Duddu, S. M. K., Mariserla, S., Ariaifar, S., Zhang, S., Zhang, S., Baumgartner, S.,
588 Goenka, S., Qiu, S., Dabral, T., Walker, T., Rao, V., Khawaja, W., Zhou, W., Ren, X., Xia, Y., Chen,
589 Y., Chen, Y.-T., Dong, Z., Ding, Z., Visin, F., Liu, G., Zhang, J., Kenealy, K., Casbon, M., Kumar,
590 R., Mesnard, T., Gleicher, Z., Brick, C., Lacombe, O., Roberts, A., Yin, Q., Sung, Y., Hoffmann,
591 R., Warkentin, T., Joulin, A., Duerig, T., and Seyedhosseini, M. (2025). EmbeddingGemma:
592 powerful and lightweight text representations. *arXiv*.

593 Vintsyuk, T. K. (1968). Speech discrimination by dynamic programming. *Cybernetics*, 4(1):52–57.

594 Viterbi, A. J. (1967). Error bounds for convolutional codes and an asymptotically optimum decoding
595 algorithm. *IEEE Transactions on Information Theory*, 13(2):260–269.

596 Wang, W., Tur, G., Zheng, J., and Ayan, N. F. (2010). Automatic disfluency removal for improving
597 spoken language translation. In *2010 IEEE International Conference on Acoustics, Speech and Signal*
598 *Processing*, pages 5214–5217, Dallas, TX, USA.

599 Ziman, K., Heusser, A. C., Fitzpatrick, P. C., Field, C. E., and Manning, J. R. (2018). Is automatic
600 speech-to-text transcription ready for use in psychological experiments? *Behavior Research*
601 *Methods*, 50:2597–2605.

602 **Acknowledgements**

603 INDIVIDUAL ACKNOWLEDGEMENTS PLACEHOLDER. Our work was supported in part by
604 NSF CAREER Award Number 2145172 to J.R.M. The content is solely the responsibility of the
605 authors and does not necessarily represent the official views of our supporting organizations.
606 The funders had no role in study design, data collection and analysis, decision to publish, or
607 preparation of the manuscript.

608 **Author contributions**

609 **Competing interests**

610 The authors declare no competing interests.